

If we mix red, green and blue colour, we get the white colour. Red, green and blue is referred to as the primary colors. Similarly cyan, magenta and yellow are referred to as the primary pigment colours. If we mix magenta and yellow we get the red colour. Hence, statement 1 is correct. Similarly if cyan and magenta colours are mixed together we get blue colour. Thus, statement 2 is also correct. If we mix cyan and yellow we get the green colour.

154. Match List-I with List-II :

List-I (Colour)	List-II (Combination)
A. Magenta	1. Green and blue
B. Teal	2. Red and blue
C. Mauve	3. Blue, green and white
D. Cyan	4. Blue, red and white

Select the correct answer using the codes given below.

	A	B	C	D
(a)	2	3	4	1
(b)	2	4	3	1
(c)	4	2	1	3
(d)	3	4	2	1

69th B.P.S.C. (Pre) 2023

Ans. (a)

The correctly matched lists are as follows :

Colour	Combination
Magenta	– Red and blue
Teal	– Blue, green and white
Mauve	– Blue, red and white
Cyan	– Green and blue

155. The number of colours contained in the sunlight is –

- (a) 3 (b) 5
(c) 6 (d) 7

U.P.P.C.S. (Mains) 2015

Ans. (d)

As we all know the sunlight is a white light and it consists of seven colours. The splitting of light into its different components (VIBGYOR) i.e. violet, indigo, blue, green, yellow, orange and red is called the dispersion of light. A prism is used to disperse light into its different components.

156. Find out the odd colour from the following options–

- (a) Green (b) Brown
(c) Red (d) Yellow

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

The brown color is an odd colour. It is because of the concept of VIBGYOR. As we know that white light is the mixture of seven colours- red, orange, yellow, green, blue, indigo and violet. Thus brown doesn't belong to this combination.

157. Consider the following phenomena :

1. Size of the Sun at dusk
 2. Colour of the Sun at dawn
 3. Moon being visible at dawn
 4. Twinkle of stars in the sky
 5. Polestar being visible in the sky
- Which of the above are optical illusions?

- (a) 1, 2 and 3 (b) 3, 4 and 5
(c) 1, 2 and 4 (d) 2, 3 and 5

I.A.S. (Pre) 2013

Ans. (c)

The size of the Sun at dusk is an optical illusion. The Sun appears smaller during noon but longer at dawn or more clearly at dusk. At dawn, the Sun is seen with other objects on the horizon and looks big. The colour of the Sun at dawn is also an optical illusion. Finally twinkling of stars is also an optical illusion caused by the refraction of light while moving through various layers of atmosphere. Moon being visible at dawn and Polestar being visible in the sky are not optical illusions.

Heat and Thermodynamics

Notes

Heat :

- Heat is that energy which transfers from one system to another system as a result of their difference in temperature.
- Its SI unit is Joule.
- Generally, calorie is also used as a unit of heat.

Temperature :

- Temperature is a thermal state of an object, which expresses the direction of flow of heat.
- Heat always transfer from a substance of higher temperature to a substance of lower temperature.

Measurement of temperature :

- Thermometer is an apparatus used for measuring the temperature.
- To measure temperature, following thermometers are in use –

(i) Celsius (Centigrade) Scale :

- A scale of temperature in which 0°C represents the melting point of ice and 100°C represents the boiling point of water.

(ii) Kelvin Scale :

- On Kelvin Scale, ice melting point is 273K while the boiling point of water is 373K.

- It is clear from above that 0°C temperature on the Celsius scale is equal to 273 K on the Kelvin scale.
- So, temperature on Kelvin scale = temperature on Celsius scale + 273
 $K = C + 273$

(iii) Fahrenheit Scale :

- On this scale melting point of ice is 32°F and boiling point of water is 212°F is mentioned.

Relation between Celsius and Fahrenheit Scale

- A Temperature °C is at the Celsius scale & °F on Fahrenheit scale, then $\frac{C}{5} = \frac{F-32}{9}$

Temperature effects on matter :

- By the increase in temperature solid state converts into liquid state and liquid state converts into a gaseous state while a decrease in temperature, gaseous form converts into liquid form and liquid converts into a solid form.

(i) Melting :

- A process in which solid converts into liquid by heating.
- The temperature at which solid matter melts and converts into a liquid at atmospheric pressure is known as its melting point. For example – ice melts at 0°C into water. So, the melting point of ice is 0°C.
- Different solids have a different melting point. For example –
 The melting point of ice – 0°C
 The melting point of wax – 63°C
 The melting point of Iron – 1535°C
- The melting point of any solid shows the extent of the force of attraction between their particles (atoms or molecules). The high melting point of solid shows that it has a high force of attraction between particles.

(ii) Boiling :

- Process in which Liquid converts into vapours fastly by heating is known as boiling.
- The temperature at which liquid boils and converts into vapours very fastly at atmospheric pressure is termed as boiling point.
- Different liquids have different boiling points.
 For example –
 Alcohol – 78°C
 Water – 100°C
 Mercury – 357°C

(iii) Condensation :

- The condensation process is the change of physical condition of matter from gas to liquid and it is the contrast phenomenon of vaporization.

(iv) Freezing :

- It is a phase transition in which a liquid turns into a solid when its temperature is lowered below its freezing point and this process is the contrast of melting.
- On cooling a liquid, the temperature at which it turns into a solid is known as its freezing point.
- The freezing point of water is 0°C.
- The freezing point of the liquid and the melting point of solid is the same. For example - water freezes at 0°C and ice melts at 0°C i.e. both points are same.

Latent Heat :

- The heat required to turn a solid into a liquid or vapour or a liquid into vapour, without change of temperature is known as latent heat.
- Latent heat is of two types –
 (i) Latent heat of melting.
 (ii) Latent heat of vaporization.

Latent Heat of Melting –

- Required energy converting for solid into liquid phase is called latent heat of melting. The value of latent heat of ice is 3.34×10^5 joule / kg.

Latent Heat of Vaporization –

- Required energy converting for liquid into vapour is called latent heat of vaporization. The value of latent heat of water is 22.5×10^5 joule / kg.
- When water turns into vapour, it absorbs the latent heat. By experiments, it has been observed that burning from steam is more unendurable painful than burning from boiling water, although both are at the same temperature i.e. at 100°C.
- Steam has more latent heat in compare to boiling water. So when steam falls on the skin it forms water after condensation, it releases 22.5×10^5 joule / kg heat more than boiling water. Due to emitting more heat by steam than boiling water, steam creates more unendurable pain.

Sublimation :

- On heating when a solid changes directly into vapour and on cooling when vapour changes directly into solid then that phenomenon is known as sublimation. Some substances having this property are Ammonium chloride, Iodine, Camphor and Nephthalene, etc.

Evaporation :

- The process in which liquid turns into vapours below its boiling point is termed as evaporation. It may occur at room temperature.
- Wet clothes dry due to evaporation of water.
- The pond water dries up by evaporation of water.
- Temperature increase enhances the rate of evaporation.
- The rate of evaporation of any liquid is maximum at its boiling point.
- In case of low humidity in the air, the evaporation rate is high and water evaporates very soon, while higher humidity in the air, results into a low rate of evaporation.

Cooling by Evaporation :

- A liquid filled in the pot when evaporates, it receives latent heat of evaporation and the pot becomes cool.
- A good example of cooling by evaporation is cooling of earthen pots.
- The earthen pitcher has many minute pores in its wall. The water diffuses out from the pores and evaporates continuously. The required heat for evaporation is taken from the water itself, it leads to a lowering of temperature in the remaining amount of water.
- The sweating process is more in the summer season. Cotton clothes are the best absorbent of water. It absorbs the sweat of our body. Evaporation of sweat provides cooling to our body.
- Synthetic clothes such as polyesters are unable to absorb water in excess quantity, so these are unable to provide cooling to our body. Due to this reason, cotton clothes are given priority in the summer season.
- The fan increases the rate of evaporation of sweat so we feel the cooling.
- In Desert Room Cooler, the cooling effect is produced due to evaporation of water. It cools more in hot and dry days because in hot days, higher temperature increases the rate of evaporation.

Question Bank

1. In an ecosystem, flow of energy declines as it passes from lower to higher trophic level. This is explained by the following :
 - (a) First law of thermodynamics
 - (b) Second law of thermodynamics
 - (c) Newton's second law
 - (d) Newton's third law

M.P. P.C.S. (Pre) 2023

Ans. (b)

In an ecosystem, flow of energy declines as it passes from lower to higher trophic level. This is explained by the second law of thermodynamics. The second law of thermodynamics states that as energy is transferred or transformed, more and more of it is wasted. That means, every activity and energy transformation is accompanied by dissipation of energy. The energy decreases from one trophic level to the next trophic level because energy is lost as metabolic heat during transfer of energy. When food energy passes from producers to herbivores to carnivores, only 10% of energy is transferred from one trophic level to another trophic level.

2. The temperature point at which solid, liquid and gaseous states may stay together is known as :

- (a) boiling point
- (b) melting point
- (c) freezing point
- (d) triple point
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

The temperature point at which solid, liquid and gaseous states may stay together is known as triple point. In thermodynamics, a substance's triple point is the unique combination of temperature and pressure at which the solid phase, liquid phase, and gaseous phase can all coexist in thermodynamic equilibrium.

By international agreement, the triple point of water has been assigned a value of 273.16 K (0.01°C, 32.018 °F) at partial vapour pressure of 611.657 pascals (6.11657 mbar; 0.0060366 atm). The triple point of water was earlier used to define the Kelvin, the base unit of thermodynamic temperature in the SI system.

3. Heat transfer that does not require a medium is called:

- (a) Radiation
- (b) Reflection
- (c) Conduction
- (d) Convection

70th B.P.S.C. (Pre) 2024

Ans. (a)

Heat transfer by radiation is the process where thermal energy is transferred through electromagnetic waves, without requiring a medium. Unlike conduction and convection, radiation does not require a medium (like air or water) to travel. This means it can transfer heat across a vacuum, like the space between the Sun and the Earth. The Sun's energy travels through the vacuum of space to warm the Earth via radiation.

4. Transfer of heat by convection can take place in

- (a) Solids and liquids
- (b) Solids and vacuum
- (c) Gases and liquids
- (d) Vacuum and gases

M.P.P.C.S. (Pre) 2017

Ans. (c)

Convection is a mode of heat transfer by actual motion of matter. It is possible only in fluids and gases. Convection can be natural and forced.

5. A liquid remains hot or cold for a long time in thermos flask because there is no loss or gain of heat by

(a) Conduction
(b) Convection and radiation
(c) Both (a) and (b)
(d) None of the above

U.P.P.C.S. (Pre) 2019

Ans. (c)

Thermos flasks have a vacuum gap between the inner chamber and the outer wall, so there is no loss or gain of heat by conduction, convection and radiation. Thus, a liquid remains hot or cold for a long time in thermos flask.

6. Steam at 100°C causes more severe burns than the water at the same temperature, because :

(a) Steam is a gas
(b) Steam provides more heat
(c) Steam is highly combustible
(d) Steam exerts more pressure

Chhattisgarh P.C.S. (Pre) 2008

Ans. (b)

Steam at 100°C causes more severe burns than the water at the same temperature because steam provides more heat. The boiling point of water is 100 degree celsius. At this temperature, the water will begin to change from the liquid state to the gaseous state. For this change to take place, additional energy is required. In fact, every gram of liquid water requires 540 calories of heat energy to convert it to steam. This is called the latent heat of vaporization or latent heat of steam. So while they both would exist at the same temperature, the steam would have a lot more heat energy due to the addition of 540 calories per gram of heat energy that has been absorbed. That is why steam burns are worse than water burns.

7. Compared to a burn due to air at 100°C, the burn due to steam at 100°C is :

(a) Less dangerous due to high latent heat of evaporation
(b) More dangerous due to high latent heat of evaporation
(c) Less dangerous due to low latent heat of evaporation
(d) More dangerous due to low latent heat of evaporation

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

See the explanation of above question.

8. The hand burns more by steam comparatively to boiling water, because –

(a) The steam has latent heat
(b) Steam sneaked into the body

(c) The steam has more power
(d) Steam is gig

U.P. P.C.S. (Pre) 1993

Ans. (a)

See the explanation of above question.

9. Consider the following statements :

1. Steam at 100°C and boiling water at 100°C contain same amount of heat.
2. Latent heat of fusion of ice is equal to the latent heat of vaporization of water.
3. In an air-conditioner, heat is extracted from the room air at the evaporator coils and is ejected out at the condenser coils.

Which of these statements is/are correct ?

(a) 1 and 2
(b) 2 and 3
(c) Only 2
(d) Only 3

I.A.S. (Pre) 2003

Ans. (d)

Steam at 100°C contains 540 calories/gm more heat than the boiling water at the same temperature due to the latent heat of vaporization. The latent heat of fusion of ice is 80 cal/gm while latent heat of vaporization of water is 540 cal/gm. Thus statement 1 and statement 2 are incorrect, while statement 3 is correct. In an air-conditioner, heat is extracted from the room air at the evaporator coil and is ejected out at the condenser coil.

10. The heat required in calories to convert one gram of ice at 0° C to steam at 100° C is approximately

(a) 80
(b) 336
(c) 720
(d) 620

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

Latent heat of Melting = 80 calorie/gram
The heat required for ice at 0°C to convert in water = 80 calorie
Required heat to enhance the temperature of water from 0°C to 100°C = 100 calorie
Latent heat of vaporization = 540 calorie/gram
Hence the energy required to convert water into vapour at 100°C = 540 calorie.
Total heat required = 80 + 100 + 540 = 720 calorie.

11. Which of the following is a correct relationship between Fahrenheit (°F) and Celsius (°C) degrees?

(a) $^{\circ}\text{C} = \frac{9}{5}^{\circ}\text{F} + 32$
(b) $^{\circ}\text{C} = ^{\circ}\text{F} - 273.15$
(c) $^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$
(d) $^{\circ}\text{F} = ^{\circ}\text{C} - 273.15$

U.P.P.C.S. (Pre) 2024

Ans. (c)

The correct relationship between Fahrenheit ($^{\circ}\text{F}$) and Celsius ($^{\circ}\text{C}$) degree :

$$^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$$

This formula is derived by understanding the difference between the freezing and boiling points of water on both scales and then establishing a proportional relationship between them.

12. At what temperature are the reading of a Centigrade and Fahrenheit thermometer the same?

- (a) -40 (b) 212
(c) 40 (d) 100

U.P.P.C.S. (Mains) 2015

U.P. U.D.A./L.D.A. (Pre) 2003

I.A.S. (Pre) 1993

Ans. (a)

-40° is the temperature, at which the readings are same in both degree Centigrade and in Fahrenheit scale. Relation between degree Centigrade & degree Fahrenheit is

$$\frac{C}{5} = \frac{F - 32}{9}$$

Placing $F = -40^{\circ}$ to verify

$$C = \frac{(-40 - 32) \times 5}{9} = \frac{-72 \times 5}{9} = -40^{\circ}$$

13. At what degree are the Centigrade and Fahrenheit temperatures the same?

- (a) 100° (b) 80°
(c) -32° (d) -40°

M.P.P.C.S. (Pre) 1994

M.P.P.C.S. (Pre) 1995

Ans. (d)

See the explanation of above question.

14. At which temperature, the Celsius and Fahrenheit scales coincides?

- (a) 0° (b) 10°
(c) -40° (d) 40°
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

See the explanation of above question.

15. At what point the Fahrenheit temperature is double of Centigrade temperature?

- (a) -6.7°C (b) -12.3°C
(c) 12.3°C (d) 160°C

R.A.S./R.T.S.(Pre) 2007

Ans. (d)

According to formula

$$C/5 = (F-32)/9 \quad [\text{placing } F = 2C]$$

$$\square \quad 9C = 10C - 160$$

$$\square \quad C = 160^{\circ}$$

Hence at the temperature of 160°C , the Fahrenheit temperature is double.

16. The body temperature of a healthy man is –

- (a) 37°Celsius (b) $37^{\circ}\text{Fahrenheit}$
(c) $98.4^{\circ}\text{Celsius}$ (d) $98.4^{\circ}\text{Kelvin}$

Uttarakhand P.C.S. (Pre) 2005

Ans. (a)

Though the body temperature measured of an individual can vary, a healthy human body can maintain a fairly consistent body temperature that is around the mark of 98.6 degree Fahrenheit or about 37.0 degree Celsius. If we convert 37 degree Celsius to Kelvin, it equals to $37 + 273 = 310$ Kelvin.

17. The normal temperature of human body on the Kelvin scale is –

- (a) 280 (b) 290
(c) 300 (d) 310

I.A.S. (Pre) 1995

Ans. (d)

See the explanation of the above question.

18. The normal temperature of human body is 98.4°F . It's equivalent temperature in $^{\circ}\text{C}$ is –

- (a) 40.16 (b) 36.89
(c) 35.72 (d) 32.36

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

The formula to convert Fahrenheit ($^{\circ}\text{F}$) to Celsius ($^{\circ}\text{C}$) is

$$^{\circ}\text{C} = \frac{(^{\circ}\text{F} - 32) \times 5}{9}$$

Since the temperature in $^{\circ}\text{F} = 98.4^{\circ}$

$$\therefore ^{\circ}\text{C} = \frac{(98.4 - 32) \times 5}{9} = 36.89^{\circ}\text{C}$$

19. Normal temperature of human body on Celsius scale will be –

- (a) 310 degree (b) 98.4 degree
(c) 36.9 degree (d) 31.5 degree

41st B.P.S.C. (Pre) 1996

Ans. (c)

See the explanation of above question.

20. Which one of the following scales of temperature does not have a negative value?

- (a) Celsius (b) Fahrenheit

(c) Kelvin

(d) Reaumur

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

The Kelvin scale has no negative values. The Kelvin scale is the best representation of average kinetic molecular energy. When a substance has a temperature of 0K (−273.15° C) it is at 'absolute zero' and has no kinetic energy at all because its particles stop moving.

21. A measured temperature in Fahrenheit scale is 200°F. What will be its value in Celsius scale?

(a) 93.3°C

(b) 40°C

(c) 99°C

(d) 30°C

U.P.B.E.O. (Pre) 2019

Ans. (a)

The formula to convert Fahrenheit (°F) to Celsius (°C) is

$$C = \frac{(F - 32) \times 5}{9}$$

Since the temperature in °F = 200°F

$$\begin{aligned} \therefore C &= \frac{(200 - 32) \times 5}{9} \\ &= \frac{168 \times 5}{9} = \frac{840}{9} = 93.3 \end{aligned}$$

So, the value of 200°F is 93.3°C in Celsius scale.

22. Let the average temperatures in Centigrade (C) and Fahrenheit (F) be $\bar{C}$ and $\bar{F}$. If $\bar{C}$ and $\bar{F}$ are related to $F = \frac{9}{5}C + 32$, then $\bar{F}$ and $\bar{C}$ have the relation :

(a) $\bar{F} = \frac{9}{5}\bar{C} + 32$

(b) $\bar{F} = \bar{C} + 32$

(c) $\bar{F} = \frac{9}{5}\bar{C}$

(d) $\bar{F} = \frac{9}{5}\bar{C} - 32$

U.P. P.C.S. (Mains) 2016

Ans. (a)

The relationship between Centigrade (C) and Fahrenheit (F) is –

$$F = \frac{9}{5}C + 32$$

∴ Average temperature in Centigrade is $\bar{C}$ and Fahrenheit is $\bar{F}$

∴ Relationship between $\bar{C}$ and $\bar{F}$

$$\bar{F} = \frac{9}{5}\bar{C} + 32$$

23. The value of 40 degrees Celsius in Fahrenheit scale is

(a) 104° F

(b) 100° F

(c) 102° F

(d) 75° F

(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

$$F = \frac{9}{5}C + 32$$

where, C = Centigrade Temperature

F = Fahrenheit Temperature

$$F = \frac{9}{5} \times 40 + 32 \Rightarrow F = 72 + 32 \Rightarrow F = 104$$

So, the value of 40° C is 104°F at Fahrenheit scale.

24. The value of 50°C in Fahrenheit scale is :

(a) 104 °F

(b) 122°F

(c) 100°F

(d) 75°F

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

$$F = \frac{9}{5}C + 32$$

$$\square F = \frac{9}{5} \times 50 + 32$$

$$\square F = 90 + 32 \square F = 122$$

So, the value of 50°C is 122°F at Fahrenheit scale.

25. In cold countries, alcohol is preferred to mercury as a thermometric liquid because :

(a) Alcohol has a lower freezing point

(b) Alcohol is a better conductor of heat

(c) Alcohol is cheaper than mercury

(d) The world production of alcohol is greater than that of mercury

U.P.P.C.S. (Pre) 1997

Ans. (a)

There are lots of reasons for using alcohol than mercury as thermometric liquids. Some alcohol has a very low freezing point of about -112°C and therefore is suitable to record very low temperature. Thus, in cold countries, alcohol is preferred to mercury as a thermometric liquid.

26. Which one of the following is used as an antifreeze for the automobile engines ?

(a) Ethanol

(b) Ethylene glycol

(c) Methanol

(d) Propyl alcohol

U.P.P.C.S. (Spl.) (Pre) 2008

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (b)

Ethylene glycol is an organic molecule most widely used as antifreeze in automobile engines and as an industrial solvent. When it is mixed with water, the freezing point of the mixture is depressed, specifically a mixture of 60% ethylene glycol and 40% water freezes at −45°C. That is why water is not frozen in the engine of cold countries.

27. What is absolute zero temperature?

- (a) The starting point of any temperature scale
- (b) Theoretically the lowest possible temperature.
- (c) The temperature at which the vapour of any liquid substance is condensed.
- (d) The temperature at which all material are in vapour form.

Jharkhand P.C.S. (Pre) 2010

Ans. (b)

Absolute zero is the lowest possible temperature at which point the atoms of a substance transmit no thermal energy. They are completely at rest. It is 0 on the Kelvin scale, which translates to -273.15 degree Celsius.

28. Expansion during heating –

- (a) Occurs only in solid
- (b) Increases the weight of the material
- (c) Decreases the density of the material
- (d) Occurs at the same rate for all liquids and solids

U.P.P.C.S. (Pre) 2006

Ans. (c)

Density is defined as the ratio between mass and volume or mass per unit volume (Density = mass / volume). As the temperature increases, most materials expand or increase their volume. This results in a decrease in density.

29. When ice melts, then–

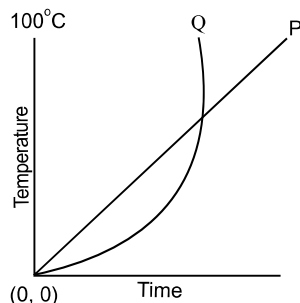
- (a) Volume increases
- (b) Volume decreases
- (c) Mass increases
- (d) Mass decreases

U.P.P.C.S. (Pre) 2012

Ans. (b)

Elements such as cast iron, ice, antimony, bismuth, and brass when melts, their volume decreases. These type of solids floats in their own melted fluid.

30. The variations in temperature from 0°C to 100°C with respect to time of two liquids P, Q are shown in the graph given below:



Which of the following statement is correct?

- (a) During heating, liquid P has always warmer than Q.
- (b) At no point of time during heating did the two liquids have the same temperature.

- (c) P attained the temperature of 100°C faster than Q.
- (d) Q attained the temperature of 100°C faster than P.

I.A.S. (Pre) 1995

Ans. (d)

In the given diagram, P denotes a straight line, while point Q denotes a curved line. It is clear that Q attained the temperature of 100°C quicker than P.

31. What is the use of thermostat?

- (a) To measure the temperature
- (b) To increase the temperature
- (c) To maintain the temperature
- (d) To convert heat into electricity

M.P.P.C.S. (Pre) 2005

Ans. (c)

Thermostat is a device that automatically regulates temperature. It helps to save energy/power.

32. Thermostat is related to :

- (a) Humidity
- (b) Temperature
- (c) Air
- (d) Clouds

U.P.P.C.S. (Pre) 1993

Ans. (b)

See the explanation of above question.

33. Thermostat is a device which :

- (a) Measures heat
- (b) Measures temperature
- (c) Automatically regulates the temperature of a system
- (d) Automatically regulates pressure of a system

U.P.P.C.S. (Pre) 2007

Ans. (c)

Thermostat is a device which automatically regulates temperature or provides a signal used by another device to regulate temperature.

34. Thermostat is a mean of keeping stable ?

- (a) Current
- (b) Temperature
- (c) Pressure
- (d) Heat Radiation

Chhattisgarh P.C.S. (Pre) 2003

Ans. (b)

See the explanation of above question.

35. What is the function of the thermostat of a refrigerator?

- (a) To lower the temperature
- (b) To increase the freezing point
- (c) To maintain constant temperature
- (d) To decrease the melting point

U.P.P.C.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

36. The safest temperature for keeping food fresh in a refrigerator is

- (a) 4°C (b) 8°C
(c) 0°C (d) 10°C

U.P.P.C.S. (Pre) 2007

Ans. (a)

Refrigerator temperatures do not destroy pathogenic or spoilage microorganisms. The lower temperature slows the growth of microorganisms already in the food. Some bacteria grow most rapidly in the range of temperature between 40° and 140°F, the "Danger Zone". Thus it is necessary to make sure that the refrigerator is maintaining the recommended temperature of 35 to 40 degree F or below. According to international standards, the recommended temperature of the refrigerator is 36°F to 38°F (1.7°C to 3.3°C). Thus option (a) is correct.

37. When soggy biscuits are kept inside the fridge for sometime they become crisp because :

- (a) Cooling releases extra moisture
(b) Humidity inside the fridge is low and extra moisture is absorbed.
(c) Humidity inside the fridge is high and extra moisture is absorbed.
(d) Pressure inside the fridge is high and help in releasing extra moisture.

U.P.P.C.S. (Pre) 2009

Ans. (b)

Most of the modern frost-free refrigerators use dehumidification process in which the moisture or water vapour or the humidity is removed from the air by keeping its dry bulb (DB) temperature constant. As water vapour or moisture condenses, it drips down into a drain, which removes water to the bottom of the refrigerator. This is the main reason for soggy biscuits turns into crisp in the refrigerator.

38. Assertion (A) : A piece of ice added to the drink, cools it.

Reason (R) : Ice takes latent heat from the drink for melting resulting in the cooling of the drink.

Select the correct answer using the codes given below:

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P. Lower Sub. (Pre) 2009

Ans. (a)

Heat transmits from hot body to cold body. Ice takes latent heat from the drink to melt and makes drink cooler.

39. Assertion (A) : Large cold storage plants use ammonia as refrigerant while domestic refrigerators use chlorofluorocarbons.

Reason (R) : Ammonia can be liquefied at ambient temperatures at low pressures.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2000

Ans. (c)

Assertion (A) is correct but reason (R) is false because ammonia can be liquefied at ambient temperature at high pressure, not low pressure.

40. Consider the following statements –

Assertion (A) : If Ice collects in the freezer, the cooling in the refrigerator is affected adversely.

Reason (R) : Ice is a poor conductor.

Select the correct answer using the codes given below:

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S (Pre) 2010

Ans. (a)

Ice collects in the freezer is the result of vapour and moisture which is expelled by the food and ice tray. It adversely affects the cooling of the refrigerator, as ice is a poor conductor.

41. When a glass rod is placed in steam, its length increases but its width :

- (a) Remains unchanged (b) Decreases
(c) Increases (d) Becomes irregular

U.P.P.C.S. (Pre) 1994

Ans. (c)

When a glass rod is placed in steam, its length and width both will increase.

42. Consider the figures of a metal ball and a metal ring given below :



Ball



Annulus

The metal ball can just pass through the hole of a metal ring formed out of a strip. When the ball is heated it gets stuck. But when the metal is heated-

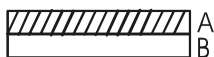
- (a) The ball can still pass through the ring as diameter expands on heating.
- (b) The ball gets stuck because of the diameter of the hole decrease on expansion.
- (c) The ball will still pass through because the hole diameter does not change.
- (d) The ball will pass through because there is no change in the ring.

I.A.S. (Pre) 1993

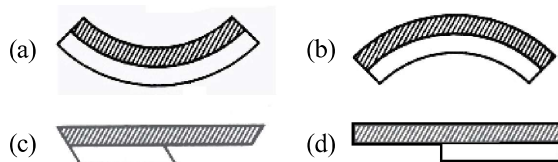
Ans. (a)

Most solids expand when heated and contract when cooled. The rate at which solids expand when heated depends on the substance. Metals tend to have higher rates of expansion than non-metal solids. When a metal ring is heated, there is an increase in outer diameter along with inner diameter. Thus, the ball can pass easily through it.

- 43. Strips of two metals A and B are firmly joined together as shown in the figure.**



On heating, A expands more than B does. If this jointed strip is heated, then it will appear as :



I.A.S. (Pre) 1999

Ans. (b)

When two strips of different metals, say one of brass and the other of iron are welded or riveted together, the greater thermal expansion of one metal results in the bending of jointed strip. Such a compound thin bar is called a bimetallic strip. When this jointed strip is heated it will bent into a curve and as metal A expands more than B, strip of metal A will be on outer (bigger) side of the curve. Hence, option (b) is the correct answer.

- 44. Assertion (A) : A piece of copper and a piece of glass are heated to the same temperature. When touched, thereafter, the copper piece appears hotter than the glass piece.**

Reason (R) : The density of copper is more than that of glass.

Code :

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not a correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

I.A.S. (Pre) 2001

Ans. (b)

Copper is a very good conductor of heat. So it allows the maximum amount of heat to pass through it from one point to other. But glass is a poor conductor of heat and hence it allows a very little amount of heat to pass through it. This is the reason that when heated to same temperature, the copper piece appears hotter than the glass piece. The density of copper is about 8920-8960 kg/m³ while the density of common glass is about 2400-2800 kg/m³. Thus, the density of copper is more than that of glass, but it is not the correct reason of the given assertion. Hence the correct answer is option (b).

- 45. In winter you touch a log of wood and a rod of iron, you will feel rod of iron is colder because –**

- (a) The heat of rod of iron is less than the heat of log of wood
- (b) In comparison to wood, iron is good conductor of heat
- (c) In comparison to wood, iron is bad conductor of heat
- (d) The rod of iron is heavier than the log of wood

41st B.P.S.C. (Pre) 1996

Ans. (b)

In winter, if we touch a log of wood and a rod of iron, we feel that rod of iron is cold. It is because of the fact that in comparison to wood, iron is a good conductor of heat. When both the metal and the wood are at the same temperature, heat will flow from iron. Similarly, when we touch wood, a poor conductor, heat does not flow from it. Hence iron rod will feel colder as compared to wood.

- 46. If the temperature of air increases, how would it affect its ability to assume water vapour –**

- (a) Decrease
- (b) Increase
- (c) Firstly decrease, then again increase
- (d) Remains constant

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

There is a direct relation between the atmospheric humidity and temperature. The ability of air to assume water vapour depends on its temperature. Thus, if temperature of air is increased then automatically its ability to hold water vapour will also be increased.

47. Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Relative humidity decreases with increasing air temperature.

Reason (R) : Absolute humidity increases with increasing evaporation.

Select the correct answer from the codes given below :

- (a) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (b) (A) is false, but (R) is true.
- (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (d) (A) is true, but (R) is false.

U.P.P.C.S. (Pre) 2024

Ans. (a)

Relative humidity expresses the air's water vapor content as a percentage of its maximum capacity at a given temperature, while Absolute humidity measures the actual amount of water vapor in the air (grams of water per cubic meter). Relative humidity is inversely related to temperature : as temperature increases, relative humidity decreases, and vice versa. Absolute humidity is not directly dependent on temperature; it simply measures the amount of water vapor present, regardless of how much the air can hold. Absolute humidity increases with increasing evaporation because evaporation is the process of adding water vapor to the air. Hence, both (A) and (R) are true, but (R) is not the correct explanation of (A).

48. In the night of strong wind, dew is not formed because

- (a) Rate of evaporation is fast
- (b) Moisture in air is low
- (c) Temperature is high
- (d) Sky is not clear

40th B.P.S.C. (Pre) 1995

Ans. (a)

Dew is not formed in the night of strong wind as due to strong wind the rate of evaporation is fast. Thus, the particles of the dew tend to evaporate.

49. Which of the following statements is not true?

- (a) Air-Conditioner and Air-Cooler both maintain temperature.
- (b) Air-Conditioner and Air-Cooler both maintain humidity
- (c) Air-Conditioner maintains humidity but Air-Cooler does not maintain humidity
- (d) Both control motion of air

U.P.P.C.S. (Pre) 2007

Ans. (b)

Air conditioner and Air-cooler both maintains temperature and the motion of air. But the Air conditioner also having the ability to maintain humidity, while air cooler doesn't have the ability to maintain humidity. Thus it is clear that option (b) is not true.

50. What does an air conditioner installed in a room control?

- (a) Temperature only
- (b) Humidity and temperature only
- (c) Pressure and temperature only
- (d) Humidity, pressure, and temperature

U.P.P.C.S. (Mains) 2011

Ans. (b)

See the explanation of above question.

51. The cooling by a desert cooler is based on –

- (a) Hot air replacement
- (b) Air dehydration
- (c) Evaporative cooling
- (d) Air rehydration

R.A.S./R.T.S.(Pre) 2012

Ans. (c)

The cooling by a desert cooler is based on evaporative cooling. These coolers are also known as swamp coolers.

52. Cooling is done by –

- (a) Flow of water
- (b) Release of compressed gas
- (c) Cooking gas
- (d) Melting the solid

47th B.P.S.C. (Pre) 2005

Ans. (b)

Most of the air conditioners use compressed gas, which can cool the room or other places.

53. In which of the following conditions wet clothes will dry earliest?

- (a) 100% RH, 60°C temperature
- (b) 100% RH, 20°C temperature
- (c) 20% RH, 20°C temperature
- (d) 20% RH, 60°C temperature

U.P.P.C.S. (Pre) 2008

Ans. (d)

The wet clothes will dry earliest at minimum humidity & maximum temperature. Thus according to options given in the question, option (d) will dry the wet clothes earliest.

54. Which one of the following changes occurs when salt is added into the water?

- (a) Boiling point is increased and freezing point decreased
- (b) Boiling point is decreased and freezing point is increased
- (c) Both boiling point and freezing point are decreased
- (d) Both boiling point and freezing point are increased

R.A.S./R.T.S.(Pre) 2008

Ans. (a)

If salt is added to water then the boiling point of water will increase. This happens as the boiling point is the temperature at which the vapor pressure of solvent becomes equal to the external atmospheric pressure. Similarly it also decreases the freezing point, which in turn will interfere with the frozen solution's crystal structure. That means the temperature will have to be colder to overcome it and freeze the mixture.

55. On adding common salt to water, the boiling point and freezing point of water will :

- (a) Increase
- (b) Decrease
- (c) Decrease and increase respectively
- (d) Increase and decrease respectively

U.P.P.C.S. (Mains) 2006

Ans. (d)

See the explanation of above question.

56. An assertion and its reason is given below -

Assertion (A) : Salt is used for cooking the food at places on higher from sea level .

Reason (R) : Temperature is low in higher places.

Which one among these is right answer ?

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

Uttarakhand P.C.S. (Pre) 2012

Ans. (b)

At high altitudes the atmospheric pressure is low. So water will boil at a lower temperature because the boiling point of water is lower at high altitude. It takes longer to cook at higher altitudes than at sea level. Adding a little salt to the water will cause the water to boil at a slightly higher temperature, which can be helpful while cooking especially at high altitudes. Lower pressure at higher altitudes causes the temperature to be colder.

57. To keep drinks at the same temperature for quite sometime, the walls of thermos bottle are coated with :

- (a) Aluminium paint
- (b) Lead Powder
- (c) Silver layer
- (d) Mercury Layer

R.A.S./R.T.S. (Pre) 1996

Ans. (c)

The glass or steel which is used in thermos bottle is coated with a silver layer to keep drinks at the same temperature for some time.

58. It is more comfortable to wear white cloth in summer because

- (a) They reflect all the heat falling on them
- (b) They radiate heat transferred from the body
- (c) They absorb perspiration
- (d) They are soothing to the eye

U.P.P.C.S. (Pre) 2007

Ans. (a)

We use white clothes in summer because white clothes are good reflectors of heat. Thus, we do not feel warm. In winter, gray or dull coloured clothes are used because gray or dull coloured clothes are good absorbers of heat.

59. White clothes are cooler than black clothes because :

- (a) They absorb whole of light
- (b) They inhibit penetration of light
- (c) They make sunlight completely cool
- (d) They reflect the whole light

U.P. R.O./A.R.O. (Mains) 2016

Ans. (d)

See the explanation of above question.

60. Assertion and Reason are given below –

Assertion (A) : We prefer the white dress in winters.

Reason (R) : White dresses are good reflectors of heat.

Which one among these is the right answer.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

Uttarakhand P.C.S. (Pre) 2012

Ans. (d)

See the explanation of above question.

61. Consider the following statements :

Assertion (A) : Light coloured clothes are preferred in summer.

Reason (R) : Light colours are poor absorbers of heat.

Select correct answer from the code given below :

- (a) Both (A) and (R) are correct and (R) correctly explains (A).
- (b) Both (A) and (R) are correct, but (R) does not correctly explain (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P. R.O./A.R.O. (Pre) 2016

Ans. (a)

White and light coloured clothes are poor absorbers of heat, that is why such clothes are preferred in summer. So both the options (A) and (R) are correct, and (R) correctly explains (A).

62. Loss of water from the body through perspiration depends on :

- (a) Atmospheric temperature only
- (b) Atmospheric humidity only
- (c) Both (a) and (b)
- (d) None of (a) and (b)

U.P. B.E.O. (Pre) 2019

Ans. (c)

Loss of water from the body through perspiration depends on both atmospheric temperature and atmospheric humidity. Perspiration or sweating allows the body to regulate its temperature. The rate of sweating depends on many factors and is increased in proportion to the work rate and the environmental temperature and humidity.

63. The most important function of perspiration is to :

- (a) Regulate body temperature
- (b) Regulate body water supply
- (c) Get rid of body poison.
- (d) Keep the skin pores clear of dirt.

R.A.S./R.T.S. (Pre) 1994-95

Ans. (a)

Perspiration is often related to heat. When there is a rise in temperature of the surroundings, the body heats up and that can be dangerous. The temperature of the body has to be stable or else the cells will start to burst. The problem will occur with the running of the organs and even heat strokes can arise which can lead to death. It is the process of perspiration that cools the body down. The sweat released onto the body cools it down and when the watery substance evaporates from the skin, it takes the heat with it giving a cooling effect.

64. A fan produces feeling of comfort during hot weather, because –

- (a) Fan supplies cold air
- (b) Our perspiration evaporates rapidly
- (c) Our body radiates more heat
- (d) Conductivity of air increase

M.P.P.C.S. (Pre) 2008

67th B.P.S.C. (Pre) 2022

Ans. (b)

In summer, a ceiling fan in the counterclockwise direction can help to cool your home by pushing cool air down. During the hot summer months, though this wind chill effect can work to your advantage by providing a breeze that can help evaporate the skin's moisture quickly, creating a cool feeling.

65. In an earthen pitcher, the water remains cold due to the process of

- (a) Condensation
- (b) Evaporation
- (c) Sublimation
- (d) None of the above

U.P.P.C.S.(Pre) 2012

Ans. (b)

The water remains cold in an earthen pitcher because of evaporation. When liquid changes to a gaseous (or vapour) state without boiling, it is known as evaporation. The earthen pitcher is made of mud and has many minute pores. It is through these pores that the water, placed inside the pitcher, oozes out. Now to evaporate, the water needs to absorb heat, which will change it to vapour. The only way the water oozing out of the pitchers can turn to vapour is by absorbing heat from the liquid within the pot. Due to this process of continuous absorption of heat from water inside the pot, in a few hours, this water becomes cool.

66. On a specific day and time, the temperature was 48°C in Churu and 24°C in Shimla. Two metallic cups identical in all respects contained water at 95°C in Churu and 71°C in Shimla. Which one of these two cups reached room temperature first?

- (a) Cup in Churu
- (b) Cup in Shimla
- (c) Both cups reached room temperature at the same time
- (d) Data are not enough to find out result

R.A.S./R.T.S.(Pre) 2008

Ans. (c)

The rate of heat loss of an object is called cooling rate
Rate of heat loss (ΔQ) $\propto$ Temperature difference (ΔT)

$$\therefore 95^\circ - 48^\circ = 47^\circ \text{ and } 71^\circ - 24^\circ = 47^\circ$$

Because of the temperature difference (ΔT) is similar in both. Thus, both cups reach the room temperature at the same time.

67. A hot body cools from 90° C to 75° C in 25 seconds. The same body in further cooling from 75° C to 60° C will take how much time?

- (a) 25 seconds
- (b) Less than 25 seconds
- (c) More than 25 seconds
- (d) Nothing can be said

U.P. P.C.S. (Mains) 2017

Ans. (c)

Newton's law of cooling states that the rate of loss of heat by a body is directly proportional to the temperature difference between the body and its surroundings.

As the temperature difference between the body and its surroundings will be lesser in the second case, so the time taken in cooling from 75°C to 60°C will be more than 25 seconds.

68. Consider the following statements regarding 'heat wave' :

1. Heat wave is considered if the maximum temperature of a station reaches at least 30°C or more for plains.
2. Heat wave is considered if the maximum temperature of a station reaches at least 40°C or more for hilly regions.

Which of the above statements is/are correct?

- (a) Both 1 and 2 (b) Only 1
(c) Only 2 (d) Neither 1 nor 2

69th B.P.S.C. (Pre) 2023

Ans. (d)

Heat wave is considered when temperature of a station reaches at least 40°C or more in plains and at least 30°C or more in hilly regions. Hence, both statements are incorrect.

69. In the season of summer, we experience a humid heat when weather :

- (a) Start of summer (b) Middle of summer
(c) Scorching weather (d) Humid weather

U.P. P.C.S. (Pre) 2003

Ans. (d)

In the season of summer, when the weather is humid, we experience humid heat.

70. Xeric condition refers to –

- (a) low temperature (b) low humidity
(c) high evaporation (d) extreme temperature

U.P.R.O/A.R.O. (Pre) 2014

Ans. (b)

Xeric condition refers to low humidity. Xeric condition is characterized by an inadequate supply of water. These conditions exist in arid habitats, extremely cold habitats, and in salt marshes.

71. The atmosphere exerts enormous pressure on us. But, we do not feel it because

- (a) Our blood exerts a pressure slightly more than that of the atmosphere.
(b) We are used to it.
(c) Our bones are very strong and can withstand the pressure.
(d) The surface area of our head is very small.

U.P.P.C.S (Pre) 2010

Ans. (a)

The atmosphere exerts enormous pressure on us but we do not feel it, because our blood exerts pressure slightly more than that of the atmosphere. We feel this pressure in water because the pressure under water is greater than our normal blood pressure.

72. Which of the following statement is not true :

- (a) Boiling point of any water solution is higher than pure water.
(b) Addition of solute to a solution increases its water potential.
(c) Water vapour pressure of any solution is lower than vapour pressure of pure water.
(d) When a solution is separated with the help of semipermeable membrane then water flow can be stopped by giving pressure to the solution.

I.A.S. (Pre) 2002

Ans. (b)

Water potential is the potential energy of water per unit volume relative to pure water in reference conditions. It quantifies the tendency of water to move from one area to another due to osmosis, gravity, mechanical pressure or matrix effects such as capillary action (which is caused by surface tension). The addition of solutes in water, lowers its water potential by which it becomes negatively charged. Adding solutes makes the water molecules less concentrated and this reduces the chemical potential of water in the system. Hence, statement of option (b) is not true while statements of other three options are correct.

73. Why does water boil at a temperature below 100°C at higher altitudes ?

- (a) The atmospheric pressure decreases and hence boiling point is lowered.
(b) The gravitational attraction is less.
(c) Because of heavy winds in mountains.
(d) None of the above is correct.

U.P.P.C.S. (Pre) 1994

Ans. (a)

At higher altitudes of a mountain, the atmospheric pressure decreases due to lack of air. Therefore, the boiling water pressure decreases and water boils below 100°C due to lowering of boiling point. The boiling point of a liquid varies depending upon the surrounding environmental pressure. A liquid in a partial vacuum has a lower boiling point than that of liquid at atmospheric pressure. A liquid at high pressure has a higher boiling point than when that liquid is at atmospheric pressure.

74. Assertion (A) : The boiling point of water decreases as the altitude increases.

Reason (R) : The atmospheric pressure increases with altitude.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

**I.A.S. (Pre) 2001
U.P.R.O./A.R.O. (Pre) 2023**

Ans. (c)

As the density of air becomes heavier near the surface of the Earth (due to gravity) and begins to lighter on higher altitudes, the atmospheric pressure decreases with increase in the altitude. The boiling point of water decreases as the altitude increases. Thus, (A) is true, but (R) is false.

75. Food gets cooked faster in a pressure cooker because –

- (a) Water starts boiling at a lower temperature due to high pressure
- (b) Water starts boiling at a higher temperature due to high pressure
- (c) Water boils only at 100°C but the heat content is higher at high pressure
- (d) Convection currents are set inside the cooker

U.P.P.C.S. (Pre) 2007

Ans. (b)

Food can be cooked faster in a pressure cooker since the boiling point of water increases with increase in pressure. As the water boils, the steam produced cannot evaporate so the pressure inside the cooker increases. The boiling point of water now becomes about 120°C. The load pin fixed on the heavy aluminum lid prevents any escape of steam. The rubber ring around the lid deals the liquid inside. Thus food gets cooked faster in a pressure cooker. Pressure cookers are more common in the mountains where elevation is high. At higher elevations, atmospheric pressure is lower and therefore water boils at lower temperature. In order to make up for that difference, people use pressure cookers.

76. Food gets cooked faster in a pressure cooker because:

- (a) water starts boiling at low temperature due to high pressure
- (b) water starts boiling at high temperature due to high pressure
- (c) water starts boiling at low temperature due to low pressure
- (d) water starts boiling at high temperature due to low pressure
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

See the explanation of above question.

77. Pressure-cooker cooks food in lesser time because

- (a) The temperature of boiling water decreases due to high pressure
- (b) Air has no effect as it is closed from all sides
- (c) Temperature of boiling water increases due to high pressure
- (d) Evaporation of the water used is very small

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

See the explanation of above question.

78. Why does food cook faster in a pressure cooker?

- (a) The increased pressure increases the boiling point.
- (b) It does not waste steam.
- (c) The food is cooked quickly by steam.
- (d) The water boils at constant temperature.

U.P. Lower Sub. (Pre) 2002

Ans. (a)

See the explanation of above question.

79. Food is cooked faster in a pressure cooker because

- (a) Heat cannot escape from the cooker.
- (b) Steam is colder than the boiling water.
- (c) Due to high pressure, the boiling point of water is raised.
- (d) Water starts boiling at a lower temperature.

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

See the explanation of above question.

80. In a pressure cooker food is cooked in shorter time because –

- (a) The boiling point of water increases
- (b) The boiling point of water decreases
- (c) The food takes less heat
- (d) None of the above.

U.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

81. Assertion (A) : The handle of pressure cooker is made up of ebonite.

Reason (R) : Ebonite is a hard material.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

Uttarakhand P.C.S. (Pre) 2012

Ans. (b)

Handle of pressure cookers is made up of ebonite because it is an insulator and prevents heat from reaching the handle and prevents burning of hands. Ebonite is a strong material. So assertion (A) and reason (R) both are correct but R is not the correct explanation of A.

82. In a pressure cooker, the temperature at which the food is cooked depends mainly upon which of the following?

1. Area of the hole in the lid
2. Temperature of the flame
3. Weight of the lid

Select the correct answer using the code given below.

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (c)

In a pressure cooker the food is cooked with the boiling of water. Boiling point is increased inside a pressure cooker by increasing the pressure. This pressure can be adjusted by changing the weight placed on the top of the cooker lid. The pressure (and cooking temperature) also depends upon area of the hole in the lid. The principle of pressure cooking is to maintain a temperature of about 120°C inside the cooker as opposed to a temperature of 100°C attained in open vessels. The weight is designed in such a way that it will kick when the pressure inside just exceeds about one atmospheric gauge thereby releasing the excess pressure. This is the pressure at which the boiling point is about 120°C . Once the requisite pressure and temperature is attained, more heat input (by increasing the temperature of the flame) results in more steam generated which is released through the lid and hence the cooking temperature remains the same, which is about 120°C . Thus, temperature at which the food is cooked does not depend upon temperature of the flame.

83. Maximum temperature inside pressure cooker depends on :

- (a) Area of hole on the top and weight kept on it
- (b) Area of hole on the top and food being cooked
- (c) Weight kept on hole at top and food being cooked
- (d) Area of hole on the top only

U.P.P.C.S. (Pre) 1995

Ans. (a)

See the explanation of above question.

84. When a bottle is filled with water and it is freezing the bottle breaks down, because –

- (a) Water expands upon freezing.
- (b) Bottle shrinks at freezing point.
- (c) The temperature outside the bottle is higher than inside.
- (d) Water expands on heating.

U.P.P.C.S. (Pre) 1995

Ans. (a)

A glass bottle containing water tightly corked breaks when it is kept in a freezer, because the frozen water will have more volume than the liquid water. So, increased volume will put pressure on the walls of the bottle and it will break.

85. A glass bottle containing water tightly corked breaks when it is kept in freezer, because –

- (a) Bottle shrinks on freezing.
- (b) The volume of water decreases on freezing
- (c) The volume of water increases on freezing.
- (d) Glass is the bad conductor of heat.

I.A.S. (Pre) 1994

Ans. (c)

See the explanation of above question.

86. When a piece of ice floating in a glass of water melts, the level of water will :

- (a) Fall
- (b) Rise
- (c) Remain same
- (d) Fall or rise depends on the temperature of water

R.A.S./R.T.S.(Pre) 2007

Ans. (c)

We know that the mass of liquid displaced by the floating ice equals to the mass of the object. That's why the ice caps float on water. Similarly, when an ice cube melts in a glass of water, the water level does not change as the melted water will occupy exactly the same amount of space as the ice under the water level previously occupied.

87. Cloudy nights are warmer compared to clear cloudless nights, because clouds–

- (a) Prevent cold waves from the sky from descending on earth.
- (b) Reflect back the heat given off by earth.
- (c) Produce heat and radiate it towards earth.
- (d) Absorb heat from the atmosphere and send it toward earth.

I.A.S. (Pre) 2001

Ans. (b)

When you have cloudy night, the clouds prevent heat from escaping through the atmosphere and into space. The clouds act like a blanket and trap the heat. That is why every time when there is a cloudy night, it is always warmer than a clear night.

88. The cloudy nights are warmer than clear nights because–

- (a) The clouds prevent radiation of heat from the ground and the air
- (b) Clouds absorb sunlight during the day and radiate the same in night

- (c) Clouds make the atmosphere damp and generate heat
(d) The clouds obstruct the movement of air which creates heat

U.P. Lower Sub. (Spl.) (Pre) 2008

Ans. (a)

The cloudy nights are warmer than clear nights because the clouds prevent radiation of heat from the ground and the air and they reflect back the heat.

89. Consider the following statements :

- High clouds primarily reflect solar radiation and cool the surface of the Earth.**
- Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause warming effect.**

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2022

Ans. (d)

Whether a given cloud will heat or cool the surface depends on several factors, including the cloud's altitude, its size, and the make-up of the particles that form the cloud. Low, thick clouds primarily reflect solar radiation and cool the surface of the Earth. While high, thin clouds primarily transmit incoming solar radiation; at the same time, they trap some of the outgoing infrared radiation emitted by the Earth and radiate it back downward, thereby warming the surface of the Earth. Hence, both of the given statements are incorrect.

90. Consider the following statements :

Statement-I : The temperature contrast between continents and oceans is greater during summer than in winter.

Statement-II : The specific heat of water is more than that of land surface.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
(c) Statement-I is correct but Statement-II is incorrect
(d) Statement-I is incorrect but Statement-II is correct

I.A.S. (Pre) 2023

Ans. (d)

The specific heat capacity is the property of the substance which determines the change in the temperature of the substance (undergoing no phase change) when a given quantity of heat is absorbed (or rejected) by it. It is defined as the amount of heat per unit mass absorbed or rejected by the substance to change its temperature by one unit. Water has the highest specific heat capacity compared to other substances. The specific heat of water is much greater than that of land surface. Owing to its high specific heat capacity, the water of ocean warms up much more slowly than the land during summer and consequently wind from the sea has a cooling effect. On the other hand, the ocean slowly cools in winter, keeping the air warm. That is the reason why ocean temperatures change relatively little from summer to winter. Thus, the temperature contrast between continents and oceans is greater during winter than in summer. Hence, Statement-I is incorrect while Statement-II is correct.

91. Fruit stored in a cold chamber exhibit longer storage life because

- (a) Exposure to sunlight is prevented
(b) Concentration of carbon dioxide in environment is increases
(c) Rate of respiration is decreased
(d) There is increase in humidity

I.A.S. (Pre) 2013

Ans. (c)

It is well known that even after harvesting of the fruits the process of respiration continues and brings about ripening, change in colour, softening of flesh, increase in sugar content and development of flavour. These changes occur more rapidly at higher temperature. This means that the lower the temperature of storage the longer would be the storage life of the fruit. In cold chambers the rate of respiration of fruits is decreased and their storage life is increased.

92. Assertion (A) : Woollen clothes keeps us warm.

Reason (R) : Woollen fibres are made from a special type of protein, which is a bad conductor of heat.

Select the correct answer from the codes given below:

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2001

Ans. (c)